

Letter & Document Template Management Module

Companion Sub-Document to Version 4.0

SaaS-Based School Management System for Liberia

April 2026

Field	Details
Document Type	Companion Module Plan (Sub-Document to v4)
Module	Letter & Document Template Management
Platform	Web-Based SaaS + PWA Mobile
Target Market	Republic of Liberia
Primary Delivery	PDF Download/Print (physical distribution)

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1. Module Overview

The Letter & Document Template Management Module provides schools with a centralized system for creating, managing, approving, and distributing official letters, notices, and documents. It integrates directly with existing system modules—Admissions, Discipline, Finance, Attendance, and Communication—to auto-populate recipient data and trigger context-aware document generation.

1.1 Purpose

Liberian schools generate hundreds of official letters each academic year—from acceptance letters to disciplinary notices to fee reminders. Currently, most schools rely on manually typed Word documents or handwritten letters, leading to inconsistency, lost records, and no audit trail. This module solves these problems by providing:

- Standardized, school-branded letter templates
- Dynamic data population from the system database
- Role-based approval workflows before sensitive letters are sent
- Multi-channel delivery suited to Liberia's infrastructure
- Complete audit trail for legal protection and accountability

1.2 Scope

This module covers all official correspondence generated by the school, organized into seven categories: Admissions, Disciplinary, Academic, Financial, Attendance, Communication, and Administrative. It does NOT cover informal messaging (handled by the Communication Hub) or WAEC-related correspondence (handled externally by WAEC).

2. Letter Categories & Types

All letter types are organized by department. Each letter type has a default template, dynamic placeholders, required approvals, and designated delivery channels.

2.1 Admissions Letters

Letter Type	Description	Triggered By	Primary Recipient
Acceptance Letter	Confirms student admission with enrollment details and fee schedule	Admin approves application	Parent/Guardian
Rejection Letter	Informs applicant of unsuccessful application with reason	Admin rejects application	Parent/Guardian
Waitlist Notification	Notifies applicant of waitlist status with expected timeline	Admin sets waitlist status	Parent/Guardian
Transfer Letter	Official release/acceptance document for inter-school transfers	Admin initiates transfer	Receiving School / Parent

2.2 Disciplinary Letters

Letter Type	Description	Triggered By	Approval Required
Warning Letter	Formal behavioral warning with incident details	Dean of Students / VP	Vice Principal
Suspension Notice	Suspension with duration, conditions for return	Dean of Students	Principal
Never To Return (NTR)	Permanent removal from school—most severe action	Principal	Principal + Proprietor review
Expulsion Letter	Formal expulsion with Ministry reporting if required	Principal	Principal + Proprietor review

2.3 Academic Letters

Letter Type	Description	Triggered By	Primary Recipient
Report Card Cover Letter	Accompanies printed report cards with term summary	End of term processing	Parent/Guardian
Promotion Notice	Confirms student advancement to next grade	Grade finalization	Parent/Guardian
Retention Notice	Informs parent of grade retention with support plan	Grade finalization	Parent/Guardian
Honor Roll Certificate	Recognizes outstanding academic achievement	GPA threshold met	Student + Parent

2.4 Financial Letters

Letter Type	Description	Triggered By	Primary Recipient
Fee Reminder	Upcoming fee deadline with amount and payment methods	Scheduled (configurable)	Parent/Guardian
Outstanding Balance Notice	Overdue balance with breakdown and consequences	Balance exceeds threshold	Parent/Guardian
Payment Receipt	Confirmation of payment received with transaction details	Payment recorded	Parent/Guardian

2.5 Attendance Letters

Letter Type	Description	Triggered By	Primary Recipient
Chronic Absenteeism Notice	Alert when absences exceed school's threshold	Absence count trigger	Parent/Guardian
Truancy Notice	Formal truancy warning with legal implications	Repeated unexcused absences	Parent/Guardian

2.6 Communication Letters

Letter Type	Description	Triggered By	Primary Recipient
PTA Meeting Invitation	Invitation with date, time, venue, and agenda	Admin creates event	All Parents
General Announcement	School-wide announcements (policy changes, events)	Admin drafts	All Stakeholders
Emergency Notice	Urgent communication (closures, safety alerts)	Admin triggers emergency	All Stakeholders

2.7 Administrative Letters

Letter Type	Description	Triggered By	Primary Recipient
Enrollment Verification	Confirms student enrollment status for third parties	Parent/admin request	Requesting Party
Recommendation Letter	Teacher/counselor recommendation for student	Teacher/Counselor drafts	External Institution
Withdrawal Confirmation	Confirms student withdrawal with clearance status	Admin processes withdrawal	Parent/Guardian

3. Template Builder Design

The Template Builder is a web-based editor that allows school administrators to create and customize letter templates. It is designed for non-technical users with a simple, intuitive interface.

3.1 Template Components

Component	Description	Customizable
School Header	School logo, name, address, contact info, motto	Yes — auto-populated from school settings
Letter Title	e.g., “Acceptance Letter”, “Suspension Notice”	Yes — editable text
Date Line	Auto-generated date or manual override	Yes — format selection
Recipient Block	Recipient name, address (auto-filled from student/parent data)	Yes — field selection
Salutation	e.g., “Dear Mr./Mrs. {{parent_last_name}},”	Yes — template with placeholders
Body Text	Main letter content with dynamic placeholders	Yes — rich text editor
Closing	Sign-off text, signatory name, title	Yes — editable
Signature Block	Digital signature image or typed name with title	Yes — upload signature image
Footer	School disclaimer, reference number, page number	Yes — editable

3.2 Rich Text Editor Features

- Bold, italic, underline formatting
- Font size and style selection (within school brand guidelines)
- Paragraph alignment (left, center, right, justify)
- Bulleted and numbered lists
- Insert dynamic placeholders from dropdown menu
- Preview mode showing populated sample data
- Copy from existing template functionality

3.3 Branding Customization

Each school's templates automatically inherit their branding from the school registration profile:

- School logo (uploaded during school setup)
- Primary and secondary colors
- School motto and tagline
- Official school address and contact information

Administrators can override these defaults on a per-template basis if needed.

4. Dynamic Placeholders System

Placeholders are variables embedded in templates that automatically populate with real data when a letter is generated. This eliminates manual data entry and ensures accuracy.

4.1 Student Placeholders

Placeholder	Source	Example Output
{{student_name}}	Student enrollment record	James K. Weah
{{student_first_name}}	Student enrollment record	James
{{student_last_name}}	Student enrollment record	Weah
{{registration_number}}	Student enrollment record	BHA-2025-0042
{{grade_level}}	Current enrollment	Grade 10
{{class_section}}	Class assignment	10-A
{{date_of_birth}}	Student enrollment record	March 15, 2009
{{enrollment_date}}	Enrollment record	September 2, 2025
{{gpa}}	Academic records	3.45

4.2 Parent/Guardian Placeholders

Placeholder	Source	Example Output
{{parent_name}}	Guardian record	Mrs. Martha Weah
{{parent_first_name}}	Guardian record	Martha
{{parent_last_name}}	Guardian record	Weah
{{parent_phone}}	Guardian record	+231 770 123 456
{{parent_email}}	Guardian record	martha.weah@email.com
{{parent_address}}	Guardian record	23 Broad Street, Monrovia

4.3 School & System Placeholders

Placeholder	Source	Example Output
{{school_name}}	School settings	Bright Horizon Academy
{{school_address}}	School settings	45 UN Drive, Monrovia, Liberia
{{school_phone}}	School settings	+231 886 555 000
{{school_email}}	School settings	info@brighthorizon.eduliberia.com
{{academic_year}}	System settings	2025–2026

{{current_term}}	System settings	First Semester
{{date}}	System (auto)	April 8, 2026
{{reference_number}}	Auto-generated	BHA-LTR-2026-0187

4.4 Context-Specific Placeholders

Placeholder	Context	Example Output
{{fee_amount}}	Financial letters	LRD 15,000
{{fee_due_date}}	Financial letters	October 15, 2025
{{outstanding_balance}}	Financial letters	LRD 8,500
{{payment_amount}}	Payment receipt	LRD 15,000
{{payment_method}}	Payment receipt	MTN Mobile Money
{{transaction_id}}	Payment receipt	TXN-MTN-20251001-4829
{{absence_count}}	Attendance letters	12 days
{{absence_threshold}}	Attendance letters	10 days
{{incident_date}}	Disciplinary letters	March 22, 2026
{{incident_description}}	Disciplinary letters	Fighting on school premises
{{suspension_start}}	Suspension notice	March 25, 2026
{{suspension_end}}	Suspension notice	April 1, 2026
{{suspension_days}}	Suspension notice	5 school days

5. Delivery Channels

Liberia's infrastructure requires a multi-channel approach. Not all families have internet access or email, so the system prioritizes physical delivery while offering digital alternatives.

5.1 Channel Overview

Channel	Priority	How It Works	Best For
PDF Download/Print	Primary	Letter generated as PDF; admin prints and distributes physically	All letter types — universal access
In-App Notification	Secondary	Letter appears in parent portal with read receipt tracking	Parents with portal access
SMS Link	Secondary	SMS with short link to view/download letter on phone	Parents with basic phones
Email Attachment	Tertiary	PDF attached to email with summary in body	Parents with email (limited in Liberia)

5.2 PDF Download/Print (Primary Channel)

This is the default and most reliable delivery method for Liberia:

- Admin generates letter → system creates formatted PDF
- PDF includes school letterhead, formatting, and signature
- Bulk print option: generate PDFs for entire class/grade
- Print queue management for schools with shared printers
- Physical distribution via students or school messengers

5.3 In-App Notification

- Letter appears as notification in parent's portal dashboard
- Read receipt: system tracks when parent opens the letter
- Parent can download PDF from the notification
- Push notification (PWA) alerts parent of new letter

5.4 SMS Link

- System sends SMS: "New letter from [School Name]. View: [short-link]"
- Short link opens mobile-optimized letter view (no login required for 24 hours)
- Secured with one-time token expiring after 24 hours
- Falls back to SMS summary if link cannot be opened

5.5 Email Attachment

- PDF attached to formatted email
- Email body contains letter summary and school contact info
- Available where parent has provided valid email address

- Lowest priority due to limited email adoption in Liberia

6. Approval Workflow

Certain letters require approval before they can be sent or printed. The approval workflow ensures accountability and prevents unauthorized communications.

6.1 Approval Matrix

Letter Type	Drafted By	Approval Required From	Approval Level
Acceptance Letter	Admin Staff	Principal	Single approval
Rejection Letter	Admin Staff	Principal	Single approval
Waitlist Notification	Admin Staff	No approval needed	Auto-send
Transfer Letter	Admin Staff	Principal	Single approval
Warning Letter	Dean of Students / VP	Vice Principal	Single approval
Suspension Notice	Dean of Students	Principal	Single approval
NTR Letter	Principal	Principal (self) + Proprietor review	Dual review
Expulsion Letter	Principal	Principal (self) + Proprietor review	Dual review
Report Card Cover	Admin Staff	No approval needed	Auto-send
Promotion Notice	System (auto)	No approval needed	Auto-send
Retention Notice	System (auto)	Principal	Single approval
Honor Roll Certificate	Admin Staff	No approval needed	Auto-send
Fee Reminder	System (auto)	No approval needed	Auto-send
Outstanding Balance	Admin Staff	No approval needed	Auto-send
Payment Receipt	System (auto)	No approval needed	Auto-send
Chronic Absenteeism	System (auto)	Dean of Students	Single approval
Truancy Notice	Dean of Students	Vice Principal	Single approval
PTA Meeting Invitation	Admin Staff	Principal	Single approval
General Announcement	Admin Staff	Principal	Single approval
Emergency Notice	Principal / VP	No approval needed	Immediate send
Enrollment Verification	Admin Staff	No approval needed	Auto-send
Recommendation Letter	Teacher / Counselor	Principal	Single approval
Withdrawal Confirmation	Admin Staff	Principal	Single approval

6.2 Approval Workflow Steps

1. Drafter creates or triggers letter generation
2. If approval required: letter enters “Pending Approval” status
3. Approver receives in-app notification with letter preview
4. Approver reviews and either Approves, Requests Changes, or Rejects
5. If approved: letter moves to “Ready to Send” queue
6. If changes requested: drafter receives feedback and revises
7. Approved letter can be printed, emailed, or sent via SMS link

6.3 Proprietor Review (NTR & Expulsion)

NTR (Never To Return) and Expulsion letters are the most consequential documents a school can issue. While the Proprietor role is read-only by design (per v4 ethics principle), these letters require a special “Proprietor Review” step:

- The Principal approves the letter
- The system automatically notifies the Proprietor
- The Proprietor can view the letter and add a comment (logged in audit trail)
- The letter is held for 24 hours before becoming sendable
- The Proprietor cannot block the letter (read-only principle) but their review is logged

This ensures the Proprietor is always informed of the most serious disciplinary actions without violating the read-only access principle.

7. Integration Points

The Letter Module connects to existing v4 system modules to pull data and trigger automated letter generation.

System Module	Integration	Auto-Triggered Letters
Admissions	Pulls applicant data, enrollment status, application decisions	Acceptance, Rejection, Waitlist, Transfer
Discipline (Guidance & Counseling)	Pulls incident records, behavioral history	Warning, Suspension, NTR, Expulsion
Finance	Pulls fee schedules, payment records, outstanding balances	Fee Reminder, Outstanding Balance, Payment Receipt
Attendance	Pulls absence counts, patterns, thresholds	Chronic Absenteeism, Truancy Notice
Academic Management	Pulls grades, GPA, promotion/retention decisions	Report Card Cover, Promotion, Retention, Honor Roll
Communication Hub	Shares notification infrastructure, delivery channels	PTA Invitation, Announcements, Emergency Notice
Student Records	Pulls enrollment data for verification and administrative letters	Enrollment Verification, Recommendation, Withdrawal

7.1 Event-Driven Triggers

Letters can be triggered automatically by system events:

- Admission decision recorded → Acceptance/Rejection letter queued
- Disciplinary incident logged → Warning letter drafted (pending approval)
- Fee deadline approaching → Fee Reminder auto-sent (7 days before)
- Payment received → Payment Receipt auto-generated
- Absence threshold exceeded → Absenteeism notice drafted (pending approval)
- Term grades finalized → Promotion/Retention notices generated
- GPA threshold met → Honor Roll certificate queued

8. Audit Trail & Compliance

Every letter generated by the system is fully logged for legal protection, accountability, and regulatory compliance.

8.1 Audit Log Fields

Field	Description	Example
Letter Reference	Unique system-generated reference number	BHA-LTR-2026-0187
Letter Type	Category and type of letter	Disciplinary — Suspension Notice
Drafted By	User who created/triggered the letter	Dean James (Dean of Students)
Draft Timestamp	When the letter was created	2026-03-22 09:15:00 WAT
Approved By	User who approved (if applicable)	Principal Sarah (Principal)
Approval Timestamp	When approval was granted	2026-03-22 11:30:00 WAT
Recipient	Student/parent the letter was sent to	Mrs. Martha Weah (Parent of James Weah)
Delivery Channel	How the letter was delivered	PDF Print + In-App Notification
Delivery Timestamp	When the letter was sent/printed	2026-03-22 14:00:00 WAT
Read Receipt	Whether the recipient opened it (digital only)	Opened: 2026-03-22 18:45:00 WAT
IP Address	IP of user who created/approved	41.86.xxx.xxx

8.2 Compliance Features

- Immutable logs: No user can delete or modify audit entries (consistent with v4 audit trail design)
- Legal hold: Letters related to NTR/Expulsion are permanently retained
- Data export: Audit logs exportable to CSV/PDF for Ministry of Education reporting
- Retention policy: All letters retained for minimum 7 years (configurable per school)
- Version history: If a letter is revised before sending, all versions are preserved

9. Starter Templates

The system ships with pre-built starter templates for every letter type. Schools can use them as-is or customize them through the Template Builder. All templates include proper school branding placeholders and dynamic data fields.

9.1 Template Example — Acceptance Letter

Below is a representative starter template showing how placeholders integrate with letter content:

[School Logo]
{{school_name}}

{{school_address}} | {{school_phone}} | {{school_email}}

Date: {{date}}

Ref: {{reference_number}}

{{parent_name}} {{parent_address}}

Dear {{parent_first_name}},

RE: Acceptance of {{student_name}} into {{school_name}}

We are pleased to inform you that {{student_name}} has been accepted into {{grade_level}} at {{school_name}} for the {{academic_year}} academic year.

Registration Number: {{registration_number}}

Please complete the enrollment process by paying the required fees of {{fee_amount}} by {{fee_due_date}}. Payment can be made via MTN Mobile Money, Orange Money, or bank transfer.

We look forward to welcoming {{student_first_name}} to our school community.

Yours sincerely,

[Signature]

Principal, {{school_name}}

9.2 Available Starter Templates

The following pre-built templates ship with every school tenant:

Category	Templates Included	Count
Admissions	Acceptance, Rejection, Waitlist, Transfer	4
Disciplinary	Warning, Suspension, NTR, Expulsion	4
Academic	Report Card Cover, Promotion, Retention, Honor Roll	4
Financial	Fee Reminder, Outstanding Balance, Payment Receipt	3
Attendance	Chronic Absenteeism, Truancy Notice	2

Communication	PTA Meeting Invitation, General Announcement, Emergency Notice	3
Administrative	Enrollment Verification, Recommendation, Withdrawal Confirmation	3
	Total Starter Templates	23

10. Role Permissions for Letters

Permission to draft, approve, and send letters is mapped to the 12 system roles defined in v4. This ensures separation of duties and prevents unauthorized correspondence.

10.1 Full Permission Matrix

Role	Draft	Approve	Send/Print	View Audit
Super Admin	All types	All types	All types	Full access
Proprietor	None (read-only)	NTR/Expulsion review only	None	Full access
Principal	Disciplinary, Academic, Communication	All types requiring approval	All approved letters	Full access
Vice Principal	Disciplinary, Communication	Warning letters	Approved letters in scope	School-level
Dean of Students	Warning, Suspension drafts	None	Approved letters in scope	Discipline only
Admin Staff	Admissions, Financial, Administrative	None	All approved letters	Own drafts
IT Admin	None	None	None	System logs only
Teacher	Recommendation letters	None	Own approved letters	Own drafts
Librarian	None	None	None	None
Guidance Counselor	Recommendation letters	None	Own approved letters	Own drafts
Student	None	None	None	View own received letters
Parent/Guardian	None	None	None	View own received letters

10.2 Key Permission Rules

- No role can both draft AND approve the same letter (separation of duties)
- Proprietor maintains read-only access consistent with v4 ethics principle
- Emergency Notices bypass approval for Principal and Vice Principal only
- System-generated letters (auto-triggered) bypass drafting but follow approval rules
- All actions are logged regardless of role

11. Technical Implementation Notes

This section provides guidance for developers implementing the Letter Module within the existing v4 architecture.

11.1 Database Schema Considerations

Table	Purpose	Key Fields
letter_templates	Stores all templates per school tenant	id, school_id, category, type, title, body_html, placeholders_used, is_starter, created_by
letter_instances	Each generated letter instance	id, template_id, school_id, recipient_id, recipient_type, status, generated_pdf_url, created_by
letter_approvals	Approval workflow tracking	id, letter_instance_id, approver_id, status, comments, timestamp
letter_deliveries	Delivery tracking per channel	id, letter_instance_id, channel, status, delivered_at, read_at, delivery_metadata
letter_audit_log	Immutable audit entries	id, letter_instance_id, action, actor_id, timestamp, ip_address, details_json

11.2 API Endpoints

Endpoint	Method	Description
/api/letters/templates	GET/POST	List and create letter templates
/api/letters/templates/:id	GET/PUT/DELETE	Manage individual template
/api/letters/generate	POST	Generate letter instance from template
/api/letters/generate/bulk	POST	Bulk generate letters for class/grade
/api/letters/:id/approve	POST	Approve or request changes on letter
/api/letters/:id/send	POST	Send letter via selected channel(s)
/api/letters/:id/pdf	GET	Download generated PDF
/api/letters/audit	GET	Query audit log with filters

11.3 Multi-Tenancy

All letter data is tenant-scoped using the same Row-Level Security (RLS) model defined in v4. Each school's templates, instances, and audit logs are completely isolated. Starter templates are cloned from a system-level seed on tenant provisioning.

End of Document

Letter & Document Template Management Module

Companion to v4 — SaaS-Based School Management System for Liberia